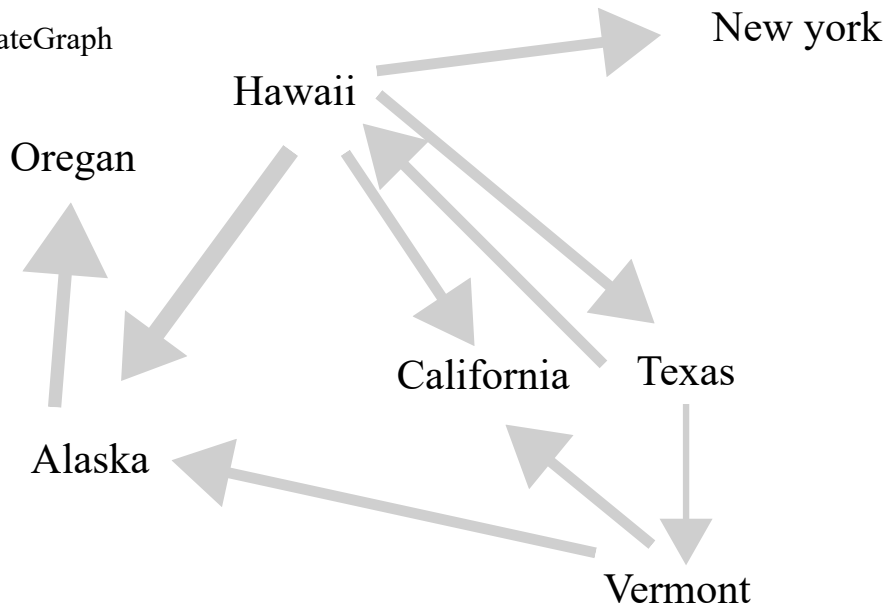


$V(\text{StateGraph}) = \{\text{Oregon, Alaska, Texas, Hawaii, Vermont, New York, California}\}$

$E(\text{StateGraph}) = \{(\text{Alaska, Oregon}), (\text{Hawaii, Alaska}), (\text{Hawaii, Texas}), (\text{Texas, Hawaii}), (\text{Hawaii, California}), (\text{Hawaii, New York}), (\text{Texas, Vermont}), (\text{Vermont, California}), (\text{Vermont, Alaska})\}$

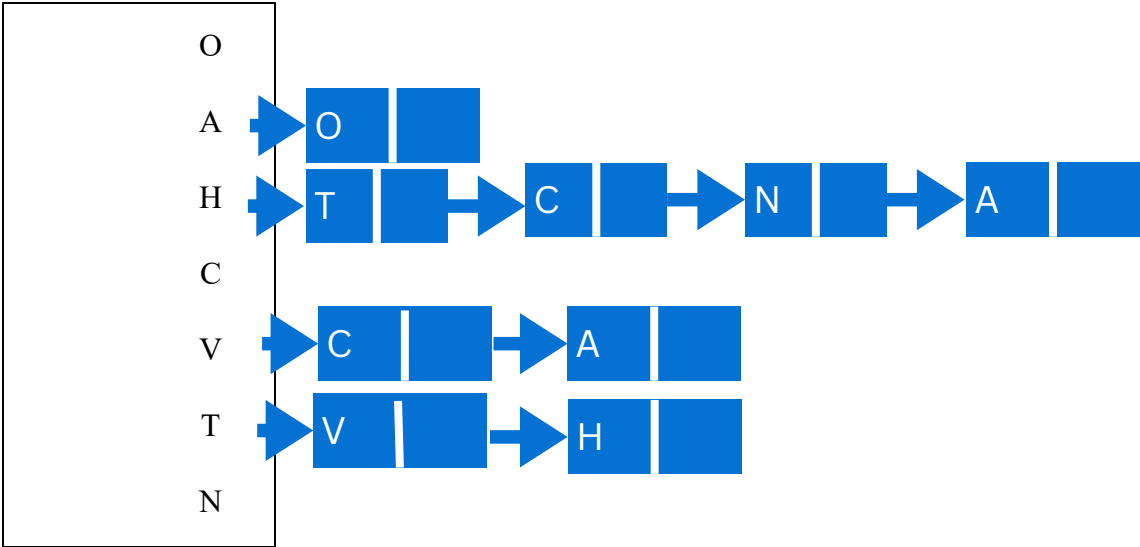
1. Draw the StateGraph

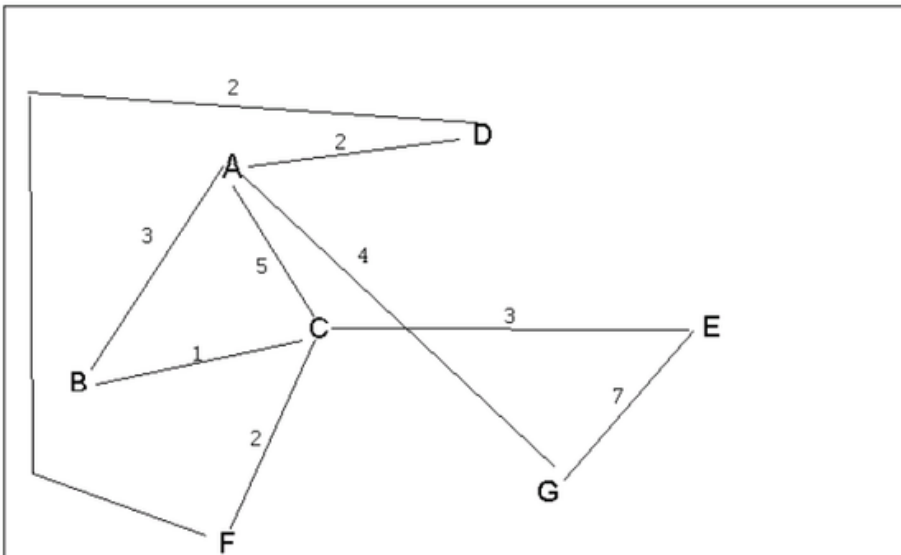


1. a. Show the adjacency matrix that would describe the edges in the graph. Store the vertices in alphabetical order

States		A	C	H	NY	O	T	V
Alaska	A	0	0	0	0	1	0	0
California	C	0	0	0	0	0	0	0
Hawaii	H	1	1	0	1	0	1	0
New York	NY	0	0	0	0	0	0	0
Oregon	O	0	0	0	0	0	0	0
Texas	T	0	0	1	0	0	0	1
Vermont	V	1	1	0	0	0	0	0

3. b. Show the adjacency lists that would describe the edges in the graph



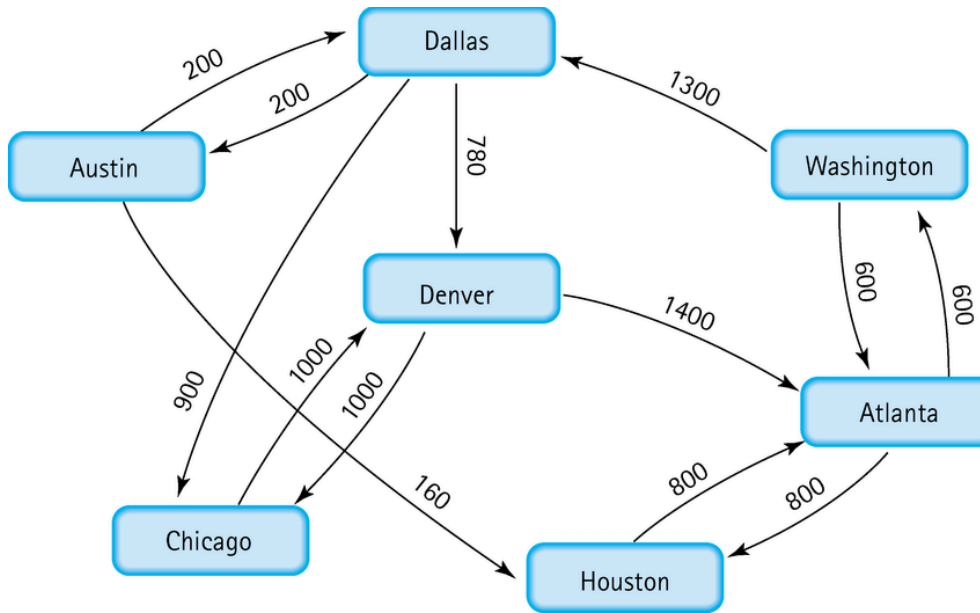


4 a. Which of the following lists the graph nodes in depth first order beginning with E?

- A) E, G, F, C, D, B, A
- B) G, A, E, C, B, F, D
- C) E, G, A, D, F, C, B
- D) E, C, F, B, A, D, G

4 b. Which of the following lists the graph nodes in breadth first order beginning at F?

- A) F, C, D, A, B, E, G
- B) F, D, C, A, B, C, G
- C) F, C, D, B, G, A, E
- D) a, b, and c are all breadth first traversals



5. Find the shortest distance from Atlanta to every other city

$At \rightarrow At = 0$

$At \rightarrow Wa = 600$

$At \rightarrow H = 800$

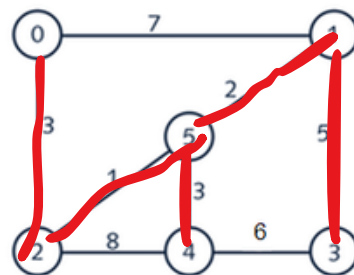
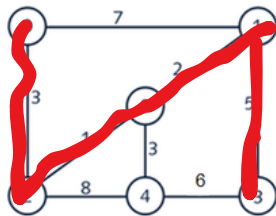
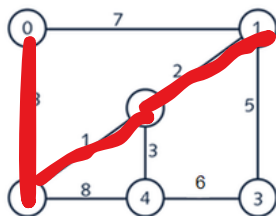
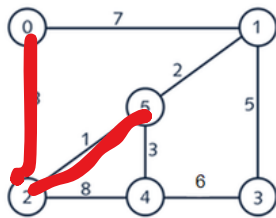
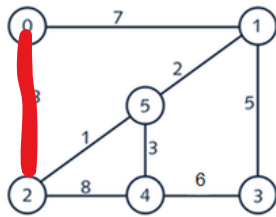
$At \rightarrow Wa \rightarrow Da = 1900$

$At \rightarrow Wa \rightarrow Da \rightarrow De = 2680$

$At \rightarrow Wa \rightarrow Da \rightarrow Au = 2100$

$At \rightarrow Wa \rightarrow Da \rightarrow C = 3000$

6. Find the minimal spanning tree using Prim's algorithm. Use 0 as the source vertex . Show the steps.



(0,2)(3)X

(0,7)(7)

(2,0)(3)

(2,5)(1)X

(2,4)(8)

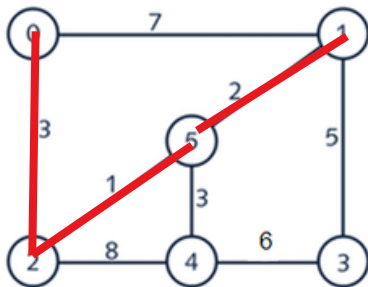
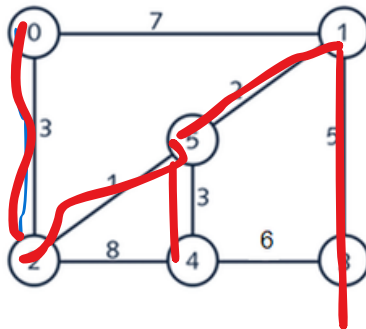
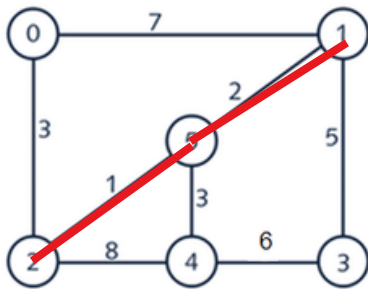
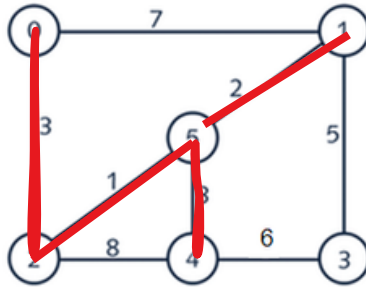
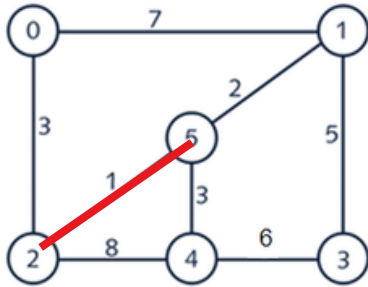
(5,1)(2)X

(5,4)(3)X

(1,3)(5)X

(4,3)(6)

7. Find the minimal spanning tree using Kruskal's algorithm. Show the weights in order and the steps.



(2,5)(1)X

(5,1)(2)X

(0,2)(3)X

(2,0)(3)

(5,4)(3)X

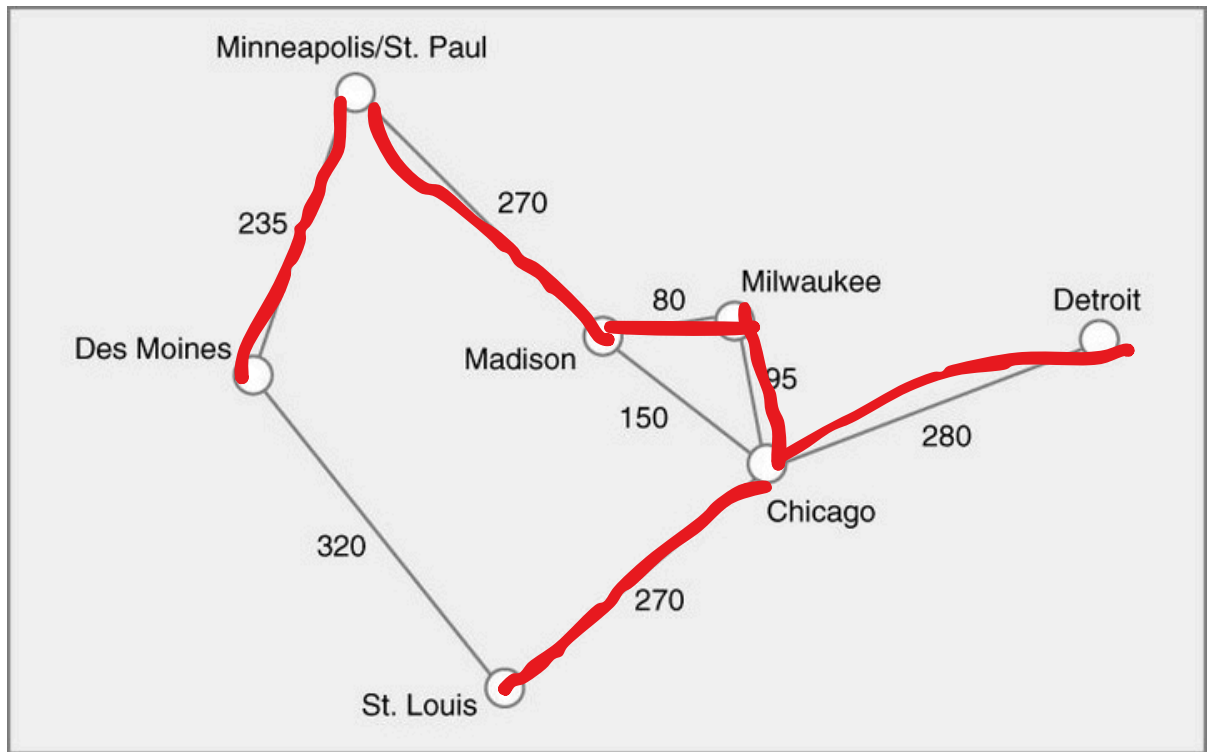
(1,3)(5)X

(4,3)(6)

(0,7)(7)

(2,4)(8)

8. Find the minimal spanning tree using the algorithm you prefer. Use Minneapolis/St. Paul as the source vertex



Prims algorithm

(Min $\rightarrow$ Des)(235)

(Min $\rightarrow$ Mad)(270)

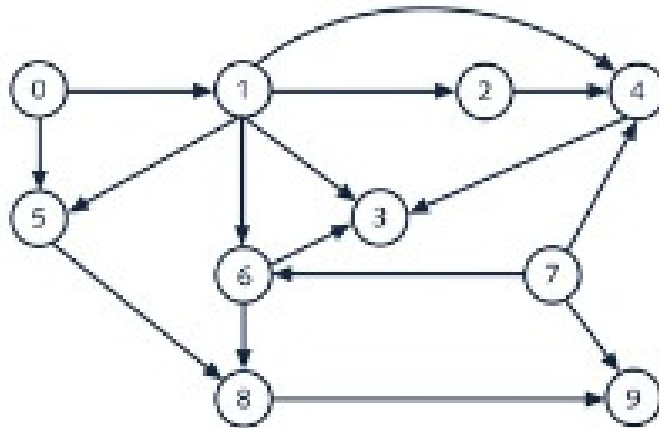
(Mad $\rightarrow$ Mil)(80)

(Mil $\rightarrow$ Ch)(95)

(Ch $\rightarrow$ St)(270)

(Ch $\rightarrow$ Det)(280)

9. List the nodes of the graph in a breadth first topological ordering. Show the steps using arrays predCount, topologicalOrder and a queue

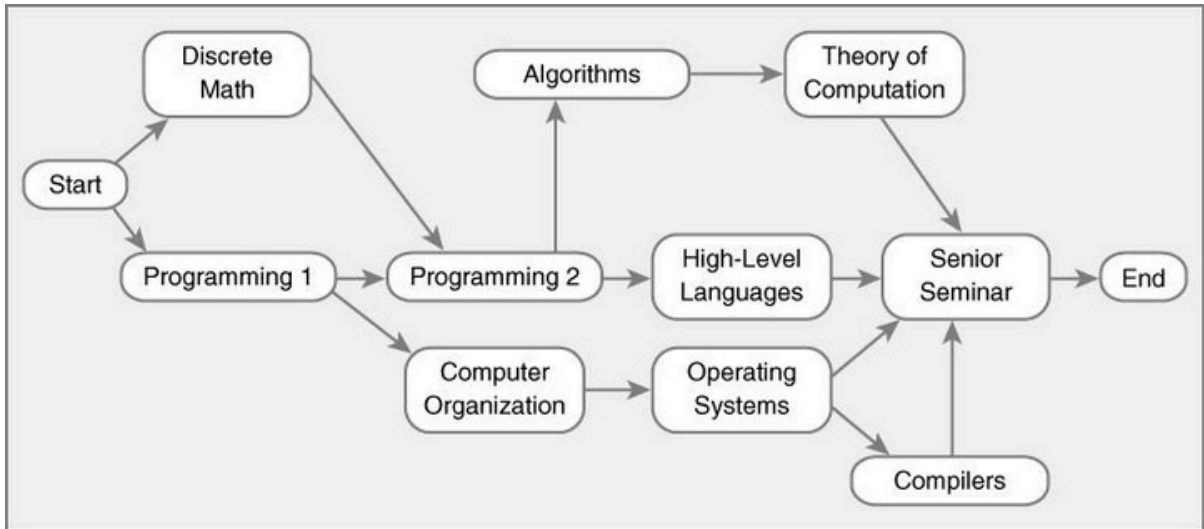


Nodes:

Pred count:	0	1	2	3	4	5	6	7	8	9
start	0	1	1	3	3	2	2	0	2	2
0	0	0	1	3	3	1	2	0	2	2
7	0	0	1	3	2	1	1	0	2	1
1	0	0	0	2	1	0	0	0	2	1
2	0	0	0	2	0	0	0	0	2	1
5	0	0	0	2	0	0	0	0	1	1
6	0	0	0	1	0	0	0	0	0	1
4	0	0	0	0	0	0	0	0	0	1
8	0	0	0	0	0	0	0	0	0	0
3	0	0	0	0	0	0	0	0	0	0
9	0	0	0	0	0	0	0	0	0	0

Topological order: 0, 7, 1, 2, 5, 6, 4, 8, 3, 9

10. List the nodes of the graph in a breadth first topological ordering.



Start

Discrete math

Programming 1

Programming 2

Computer Organization

Algorithms

High Level languages

Operating systems

Theory of computation

Compilers

Senior Seminar

End